

FINATRADES

Trade Finance Module

FinaBridge | LC | WR | POL

Implementation Options & Architecture

AI-Powered Document Verification | 5-Layer Security

Deal Room Architecture | Revenue Model

Prepared by: Finatrades Technical Architecture Team

RAMINVEST Holding Ltd. | DIFC, Dubai

Swiss FINMA Regulated Entity

March 2026 | Version 2.0

Classification: CONFIDENTIAL

. Table of Contents

1 Executive Summary

2 Platform Architecture

3 Payment Type Options

3.1 FinaBridge Wallet (Default)

3.2 Letter of Credit (LC)

3.3 Port of Loading Documents (POL)

3.4 Warehouse Receipt (WR)

4 Complete Trade Lifecycle

5 AI Document Verification Pipeline

6 Internal Review Process (3-Tier)

7 Proposal & Deal Room Flow

8 Security Architecture (5-Layer)

9 Revenue Model

10 Our Role: Pure Facilitator

1. Executive Summary

Finatrades is a gold-backed digital trade finance platform operating as a pure intermediary and facilitator. The platform does not hold gold directly, execute trades, or assume counterparty risk. Instead, Finatrades provides the technology infrastructure, AI-powered verification systems, and operational support that enables secure, transparent international trade transactions.

Gold serves as the fundamental source of truth across all Finatrades operations. While fiat currencies (USD, EUR, AED) are used for reference and settlement convenience, every transaction is ultimately backed by and denominated in gold value. This approach provides a stable, universally accepted foundation that transcends currency volatility and geopolitical risks.

Regulatory Framework

- Swiss FINMA regulated entity ensuring the highest standards of financial compliance
- RAMINVEST Holding Ltd., DIFC (Dubai International Financial Centre)
- Full adherence to international AML/KYC/CFT regulations
- OFAC, EU, and UN sanctions screening on every transaction
- Regular compliance audits and reporting

Core Value Proposition

- AI-powered document verification eliminates fraudulent submissions
- 5-layer security architecture ensures comprehensive risk management
- Deal Room provides centralized trade management for all parties
- Gold-backed instruments offer stability beyond fiat currencies
- Pure facilitator model means zero counterparty risk exposure

2. Platform Architecture

The Finatrades ecosystem consists of four interconnected modules, each serving a specific function within the trade finance lifecycle. These modules work together to provide end-to-end trade facilitation from gold custody to final settlement.

2.1 FinaVault - Gold Custody & Certification

- Physical and digital gold storage through certified vault partners
- Real-time gold certificate issuance and verification
- Master ledger maintaining immutable records of all gold positions
- Purity verification and assay certificate management
- Integration with LBMA-accredited refineries

2.2 FinaPay - Payment Infrastructure

- **MPGW (Merchant Payment Gateway Wallet):** Business-to-business payment processing and settlement
- **FPGW (Fiat Payment Gateway Wallet):** Fiat currency on/off ramp for gold transactions
- Send, receive, request, and buy gold through unified interface
- Multi-currency support with real-time gold price conversion
- Instant settlement between FinaPay wallets

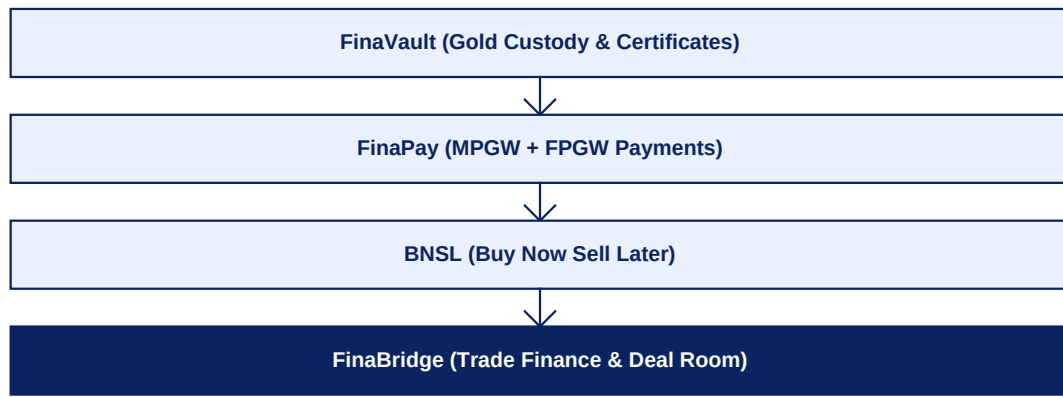
2.3 BNSL - Buy Now Sell Later

- Gold monetization framework allowing deferred sale arrangements
- Quarterly payout structures for gold holders
- Flexible holding periods with market-linked returns
- Automated settlement and distribution engine

2.4 FinaBridge - Trade Finance

- Gold-backed alternative to traditional Letters of Credit
- Support for LC, Warehouse Receipt, and Port of Loading documents
- AI-powered document verification and fraud detection
- Deal Room for centralized trade management
- Multi-party coordination between importers, exporters, and logistics

Platform Module Architecture

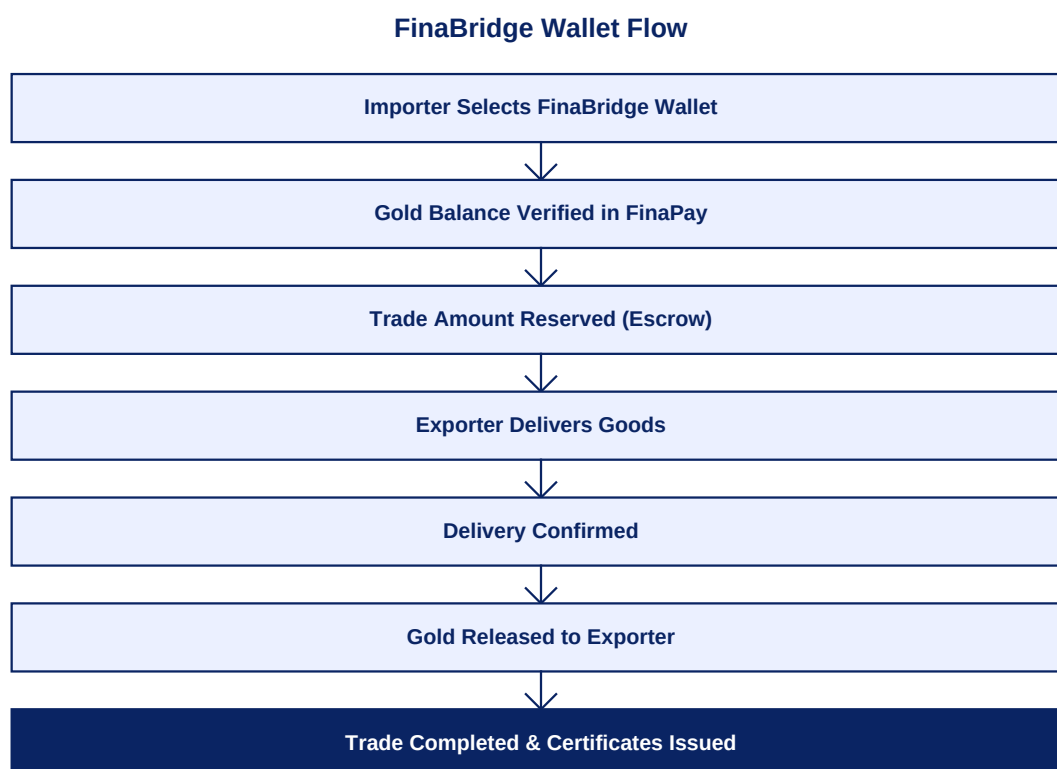


3. Payment Type Options

When an importer submits a trade request, they select a payment type that determines the trade finance instrument used. The following sections detail all available options for each payment type, including implementation alternatives and the currently active configuration.

3.1 FinaBridge Wallet (Default)

The FinaBridge Wallet is the default payment method for all trade submissions. It enables direct gold-backed wallet-to-wallet payments without requiring external financial instruments. Transactions are settled instantly within the Finatrades ecosystem using gold-denominated balances.



3.2 Letter of Credit (LC)

Letters of Credit provide importers with bank-guaranteed payment assurance to exporters. Finatrades offers four implementation approaches for LC processing, ranging from fully manual to automated. The platform currently operates under Option D, the Facilitator Model.

Option A: Manual Review

AVAILABLE

Traditional manual processing where administrators verify each LC document directly with the issuing bank through phone calls, emails, and manual cross-referencing. This approach offers maximum human oversight but lower throughput.

- User uploads LC document to the platform
- Admin manually contacts issuing bank for verification
- Bank confirms LC authenticity via phone/email/SWIFT
- Admin manually enters verified data into the system
- Credit extended to importer upon successful verification
- **Credit Limit:** Up to 90% of LC face value
- **Processing Time:** 3-7 business days

Option A: Manual LC Flow



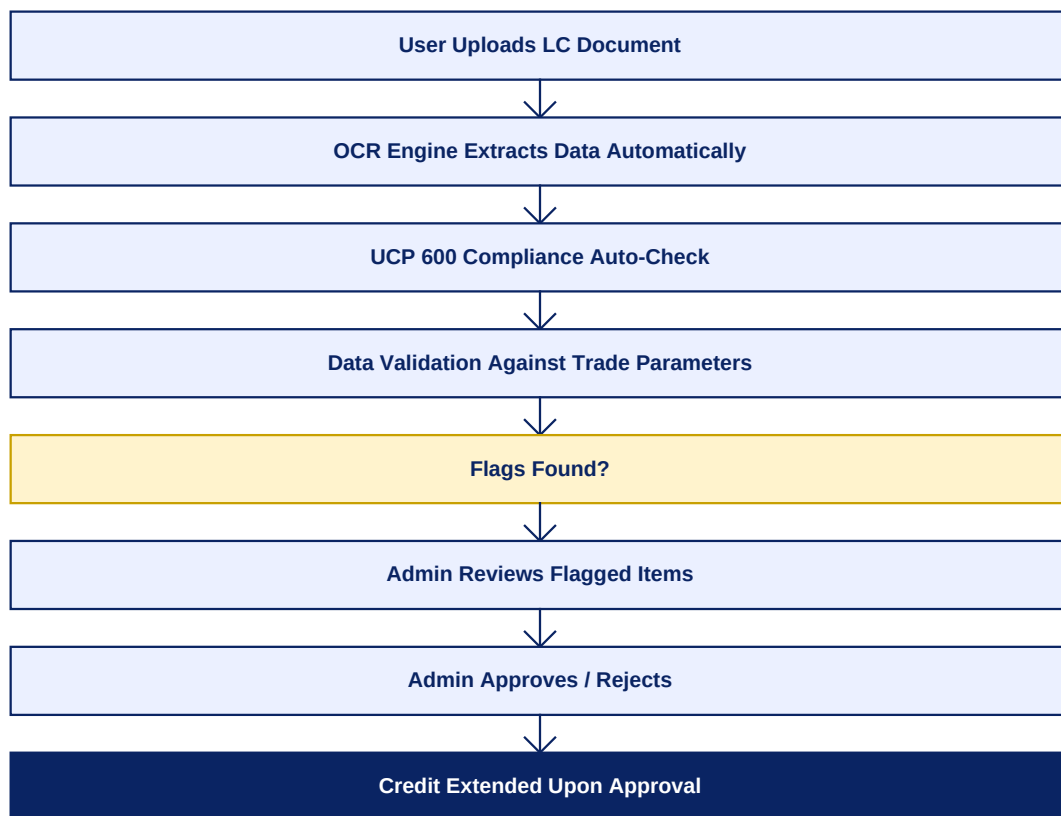
Option B: Semi-Automated

AVAILABLE

A hybrid approach combining AI-powered OCR data extraction with automated UCP 600 compliance checking, followed by human administrator approval. This reduces processing time while maintaining human oversight for final decisions.

- User uploads LC document to the platform
- OCR engine automatically extracts all data fields
- System auto-checks UCP 600 compliance rules
- Automated data validation against trade parameters
- Flagged items presented to admin for review
- Admin provides final approval or rejection
- **Credit Limit:** Up to 90% of LC face value
- **Processing Time:** 1-3 business days

Option B: Semi-Automated LC Flow



Option C: Full Automated

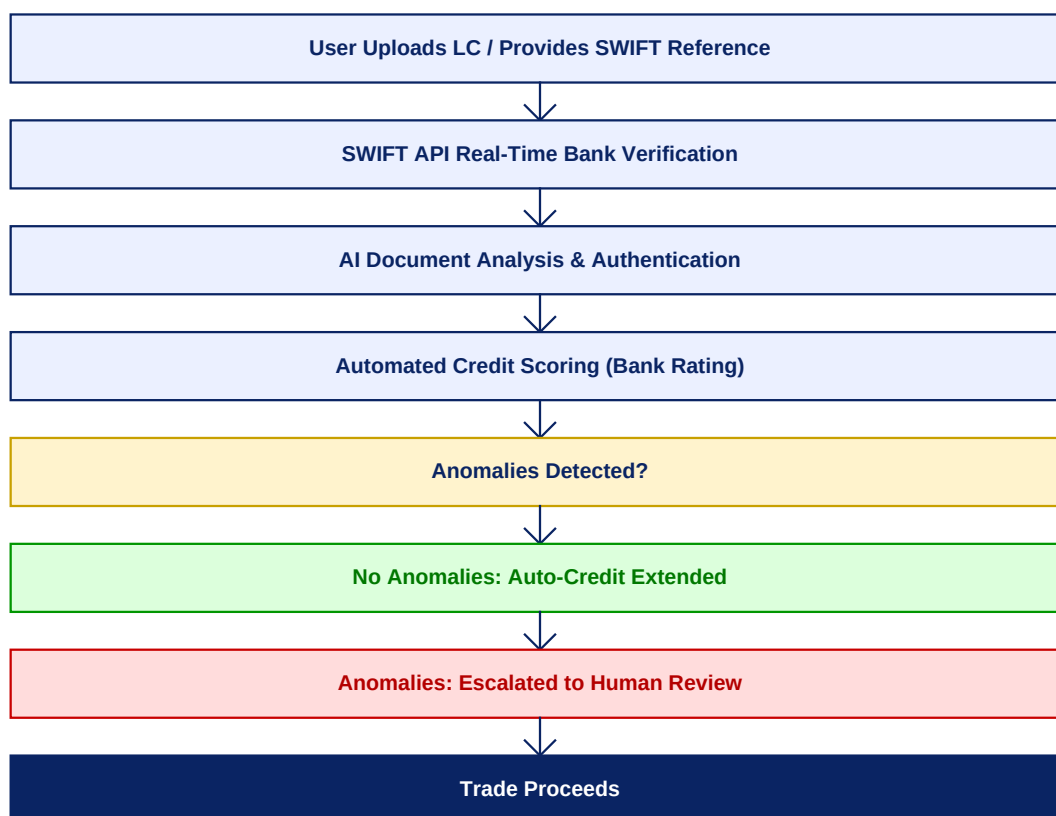
AVAILABLE

Fully automated LC processing using direct SWIFT network integration for real-time bank verification, combined with

AI-powered document analysis and automatic credit extension. Minimal human intervention required except for edge cases.

- User uploads LC document or provides SWIFT reference
- System connects to SWIFT network for real-time verification
- AI analyzes document authenticity, content, and compliance
- Automated credit scoring based on issuing bank rating
- Credit extended automatically upon successful verification
- Human review triggered only for anomalies or high-value transactions
- **Credit Limit:** Up to 90% of LC face value
- **Processing Time:** Minutes to hours (automated)

Option C: Full Automated LC Flow



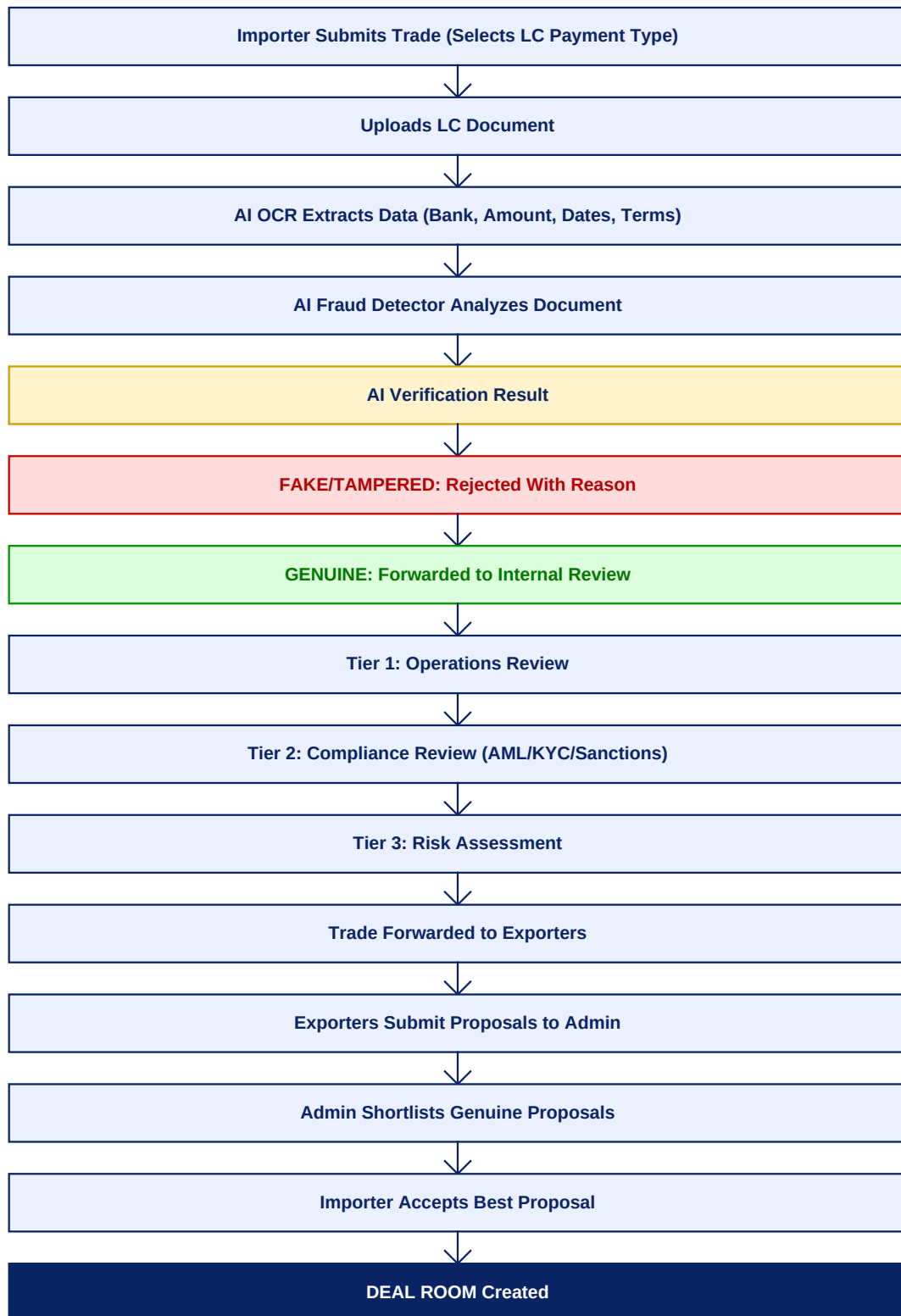
Option D: Facilitator Model

IMPLEMENTED

The currently implemented approach where Finatrades acts as a pure facilitator. The importer selects LC as the payment type and uploads the document. The AI-powered OCR engine extracts data while the Fraud Detection system verifies authenticity. Upon passing AI verification, the document proceeds through the 3-tier internal review process before the trade is forwarded to exporters for proposal submission.

- Importer selects LC as payment type when submitting trade
- Importer uploads LC document to the platform
- AI OCR engine extracts all data (issuing bank, amount, dates, parties, terms)
- AI Fraud Detector checks for forgery, tampering, inconsistencies, duplicates
- Fake/tampered documents are rejected instantly with detailed reasons
- Genuine documents forwarded to Finatrades 3-tier review team
- Operations, Compliance, and Risk teams review sequentially
- Approved trade forwarded to exporters for proposals
- **Revenue:** 1-3% of LC value + 12-18% APR on credit facility

Option D: Facilitator Model LC Flow (IMPLEMENTED)



3.3 Port of Loading Documents (POL)

Port of Loading documents (Bills of Lading, Packing Lists, Certificates of Origin) provide proof that goods have been shipped and are in transit. These documents serve as collateral for trade finance, enabling credit extension based on goods already in the shipping pipeline.

Option A: Manual Review

AVAILABLE

Traditional manual processing where administrators verify shipping documents directly with the shipping company, port authority, and freight forwarder.

- User uploads Bill of Lading and packing list
- Admin contacts shipping company for verification
- Port authority confirmation obtained manually
- Freight forwarder details cross-referenced
- Admin manually validates all document details
- **Credit Limit:** 70-80% of shipment value
- **Processing Time:** 3-5 business days

Option A: Manual POL Flow



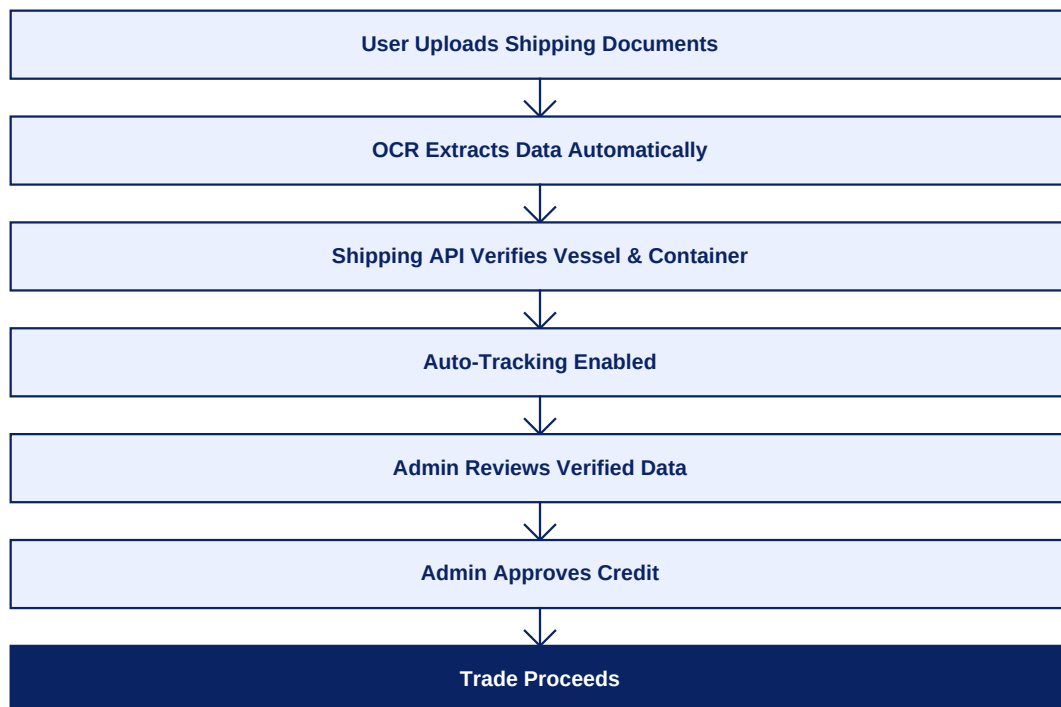
Option B: Semi-Automated

AVAILABLE

Hybrid approach using shipping line API integration for automatic vessel tracking and container status verification, combined with admin approval for final credit decision.

- User uploads shipping documents
- System auto-extracts data via OCR
- Shipping API verifies vessel, container, and route
- Auto-tracking enabled for shipment progress
- Admin reviews system-verified data and approves
- **Credit Limit:** 70-80% of shipment value
- **Processing Time:** 1-2 business days

Option B: Semi-Automated POL Flow



Option C: Full Automated

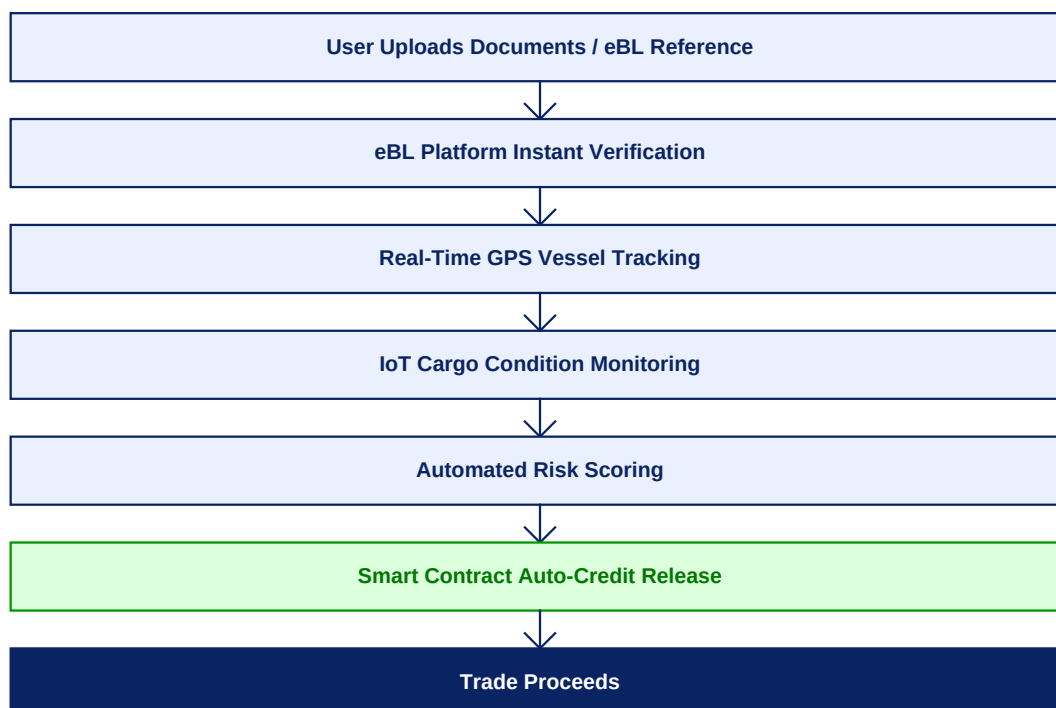
AVAILABLE

Fully automated POL processing with electronic Bill of Lading (eBL) support, real-time GPS tracking integration, smart contract-based auto-release, and automated credit scoring based on shipping data.

- User uploads documents or provides eBL reference
- eBL platform integration for instant verification

- Real-time GPS tracking of vessel and cargo
- Smart contract triggers auto-credit on verified shipment
- IoT sensors monitor cargo condition (temperature, humidity)
- Automated risk scoring based on route, carrier, and history
- **Credit Limit:** 70-80% of shipment value
- **Processing Time:** Minutes (automated)

Option C: Full Automated POL Flow



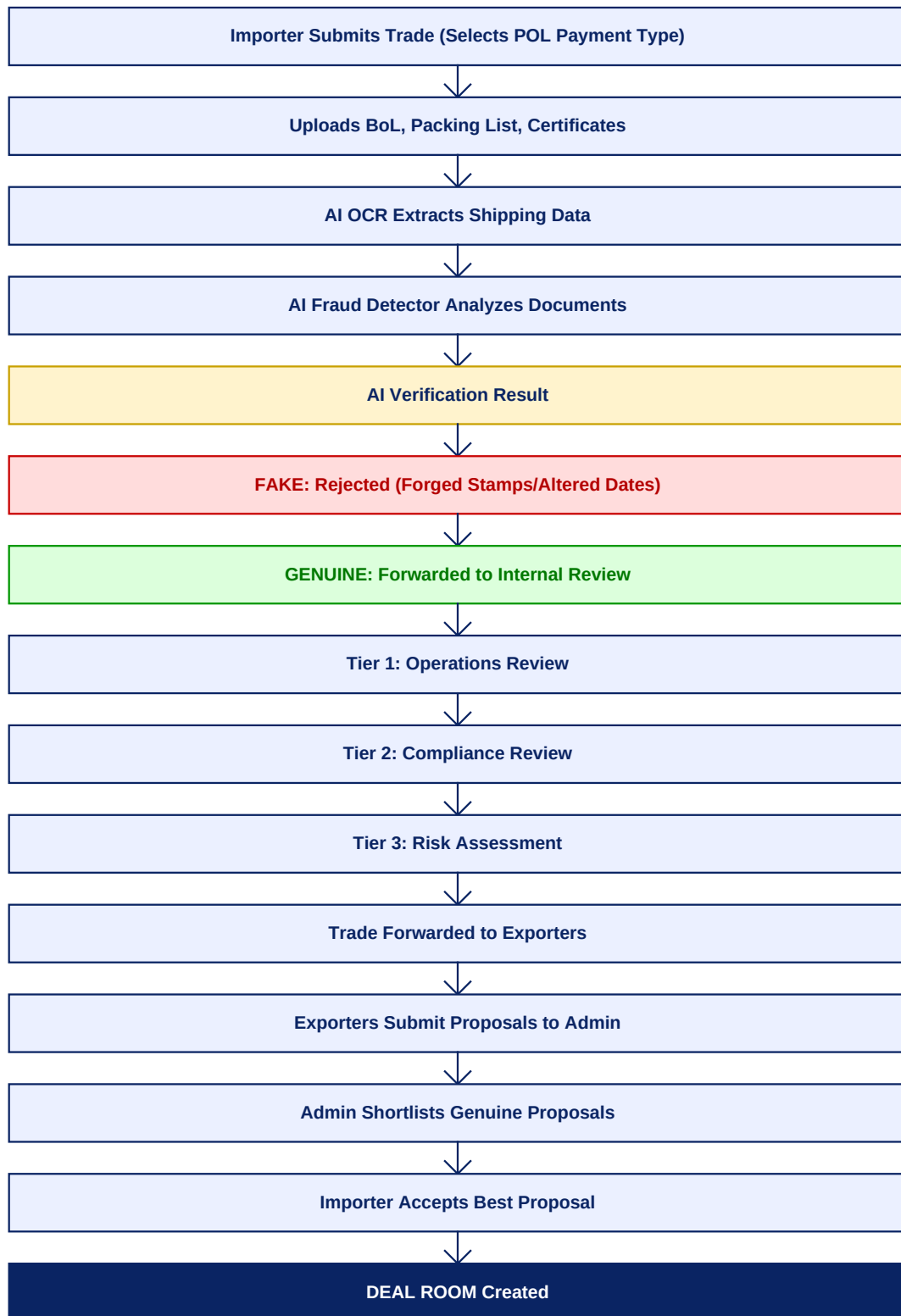
Option D: Facilitator Model

IMPLEMENTED

The currently implemented approach where Finatrades facilitates POL-based trade finance. The importer selects POL as the payment type, uploads shipping documents, and the AI system handles verification before forwarding to the review pipeline.

- Importer selects POL as payment type when submitting trade
- Uploads Bill of Lading, packing list, certificates of origin
- AI OCR extracts all shipping data (vessel, port, cargo, dates)
- AI Fraud Detector checks for forged shipping stamps, altered dates, fake carriers
- Fake documents rejected instantly; genuine forwarded to review
- 3-tier review: Operations, Compliance, Risk Assessment
- Approved trade forwarded to exporters for proposal submission
- **Revenue:** 1.5-3% transaction fee + credit interest

Option D: Facilitator Model POL Flow (IMPLEMENTED)



3.4 Warehouse Receipt (WR)

Warehouse Receipts serve as proof of commodity storage in certified warehouses, enabling credit extension against stored goods (typically gold, precious metals, or other high-value commodities). The WR acts as collateral for the trade finance facility.

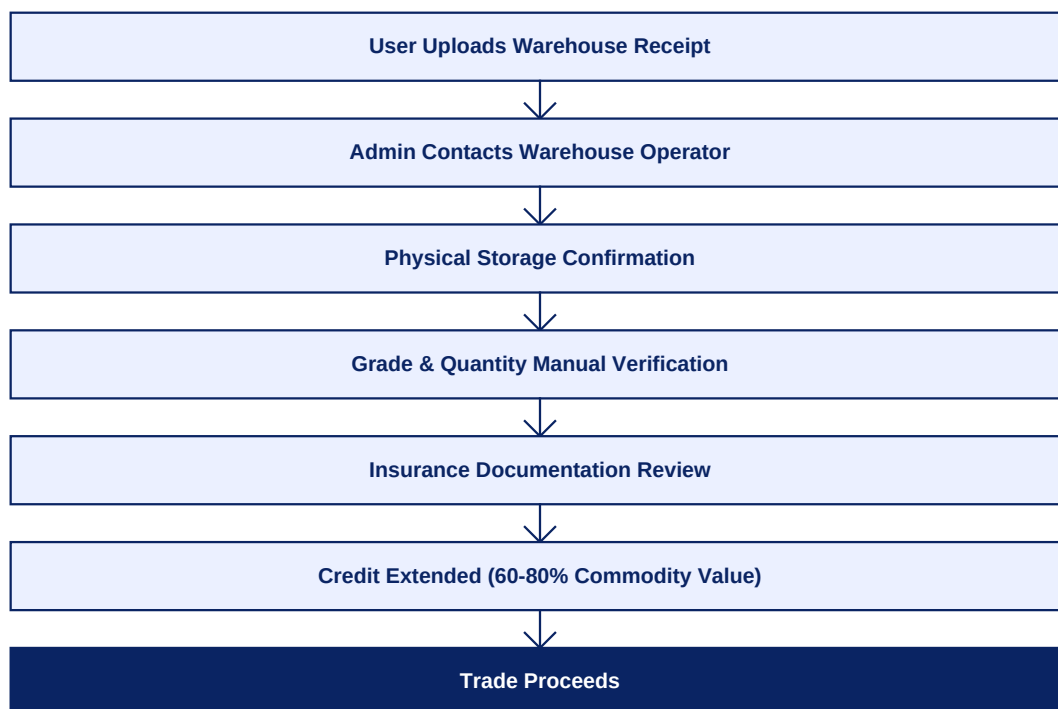
Option A: Manual Review

AVAILABLE

Traditional manual processing where administrators verify warehouse receipts directly with the warehouse operator and inspect storage documentation.

- User uploads Warehouse Receipt
- Admin contacts warehouse operator for verification
- Physical storage confirmation obtained
- Commodity grade and quantity verified manually
- Insurance documentation reviewed
- **Credit Limit:** 60-80% of stored commodity value
- **Processing Time:** 3-7 business days

Option A: Manual WR Flow



Option B: Semi-Automated

AVAILABLE

Hybrid approach utilizing a partner warehouse network for automated verification, real-time commodity valuation, and admin approval for final credit decisions.

- User uploads Warehouse Receipt
- OCR extracts receipt data automatically
- Partner warehouse network provides digital verification
- Auto-valuation based on current market prices
- Admin reviews system-verified data and approves
- **Credit Limit:** 60-80% of stored commodity value
- **Processing Time:** 1-3 business days

Option B: Semi-Automated WR Flow



Option C: Full Automated

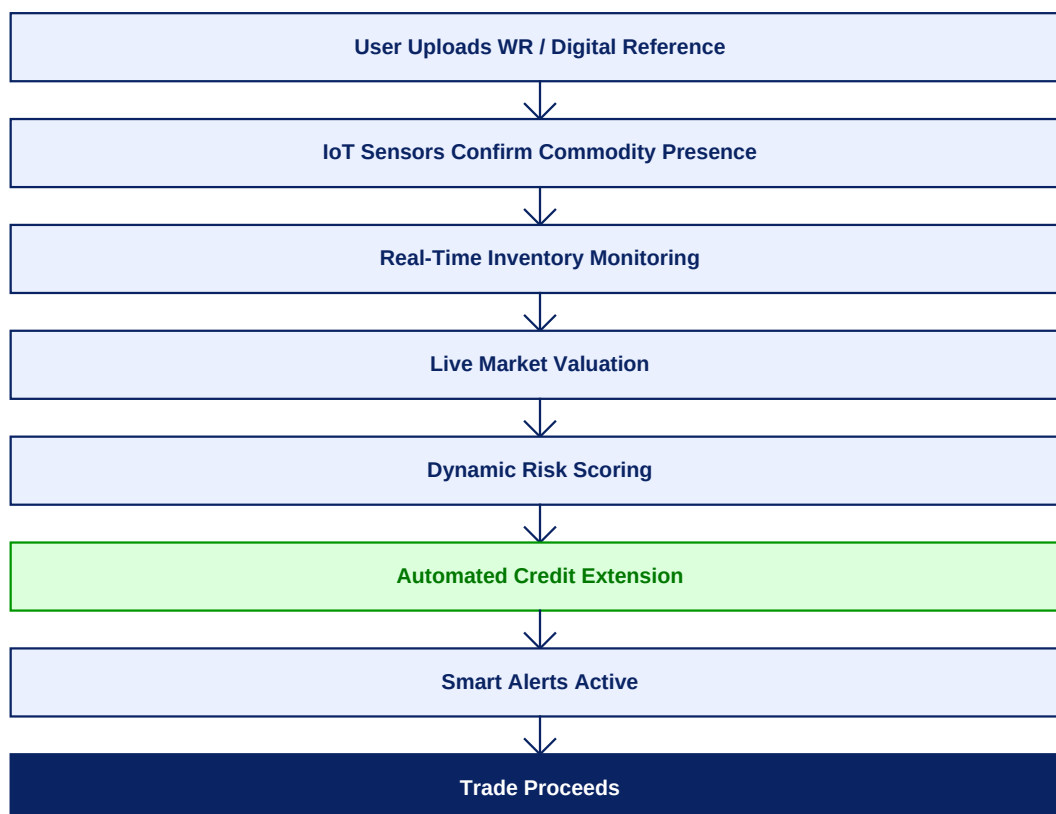
AVAILABLE

Fully automated WR processing with IoT sensor integration for real-time inventory monitoring, automated credit extension based on live commodity valuations, and dynamic risk scoring.

- User uploads WR or provides digital warehouse reference
- IoT sensors confirm commodity presence and condition

- Real-time inventory levels monitored continuously
- Automated credit based on live market valuation
- Dynamic risk scoring adjusts credit limits
- Smart alerts for inventory changes or anomalies
- **Credit Limit:** 60-80% of stored commodity value
- **Processing Time:** Minutes (automated)

Option C: Full Automated WR Flow



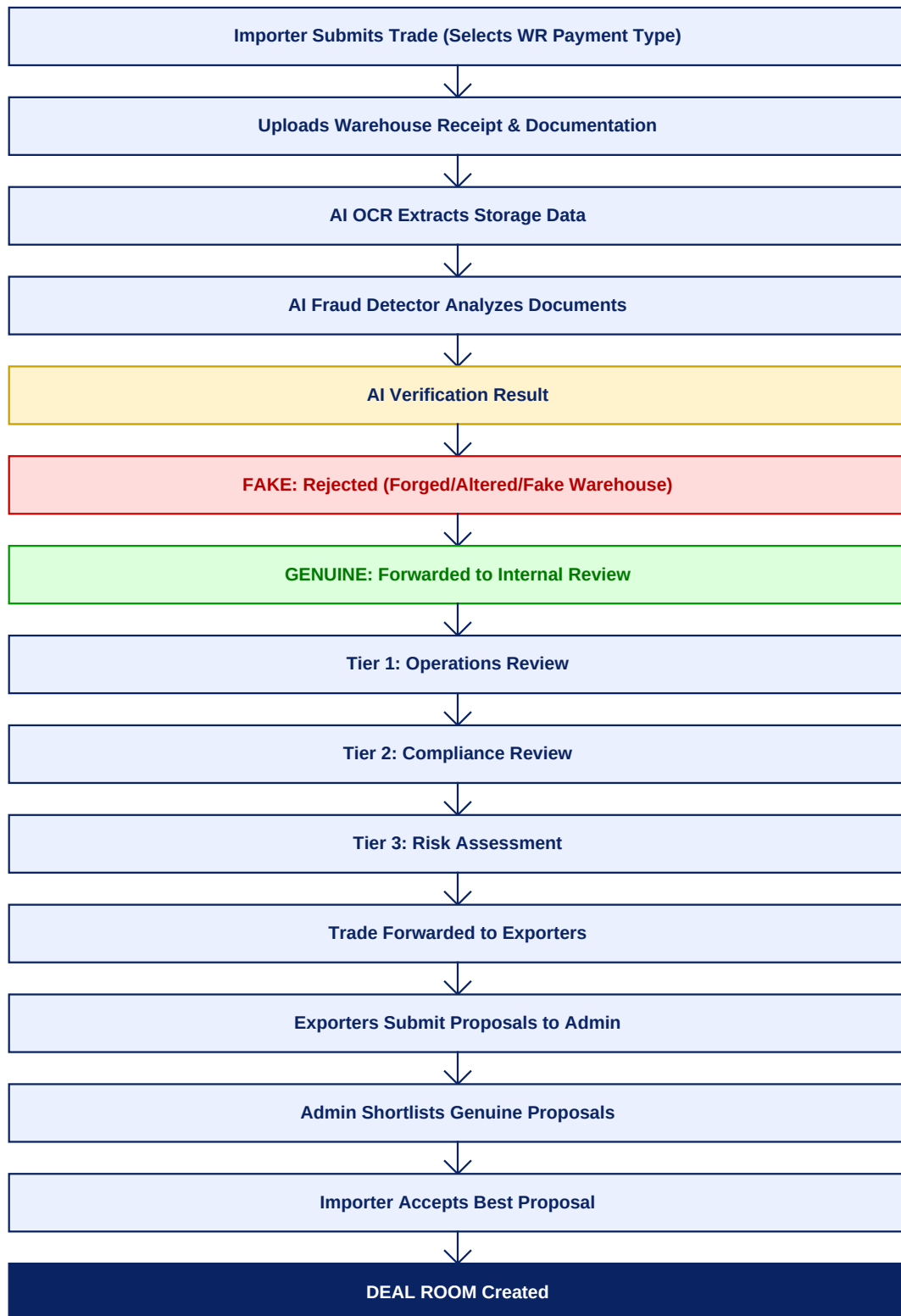
Option D: Facilitator Model

IMPLEMENTED

The currently implemented approach where Finatrades facilitates WR-based trade finance. The importer selects WR as the payment type, uploads the warehouse receipt, and the AI system handles verification before forwarding to the review pipeline.

- Importer selects WR as payment type when submitting trade
- Uploads Warehouse Receipt and supporting storage documentation
- AI OCR extracts all data (warehouse, commodity, quantity, grade, dates)
- AI Fraud Detector checks for forged receipts, altered quantities, fake warehouses
- Fake documents rejected instantly; genuine forwarded to review
- 3-tier review: Operations, Compliance, Risk Assessment
- Approved trade forwarded to exporters for proposal submission
- **Revenue:** 2-4% interest spread + storage facilitation fees

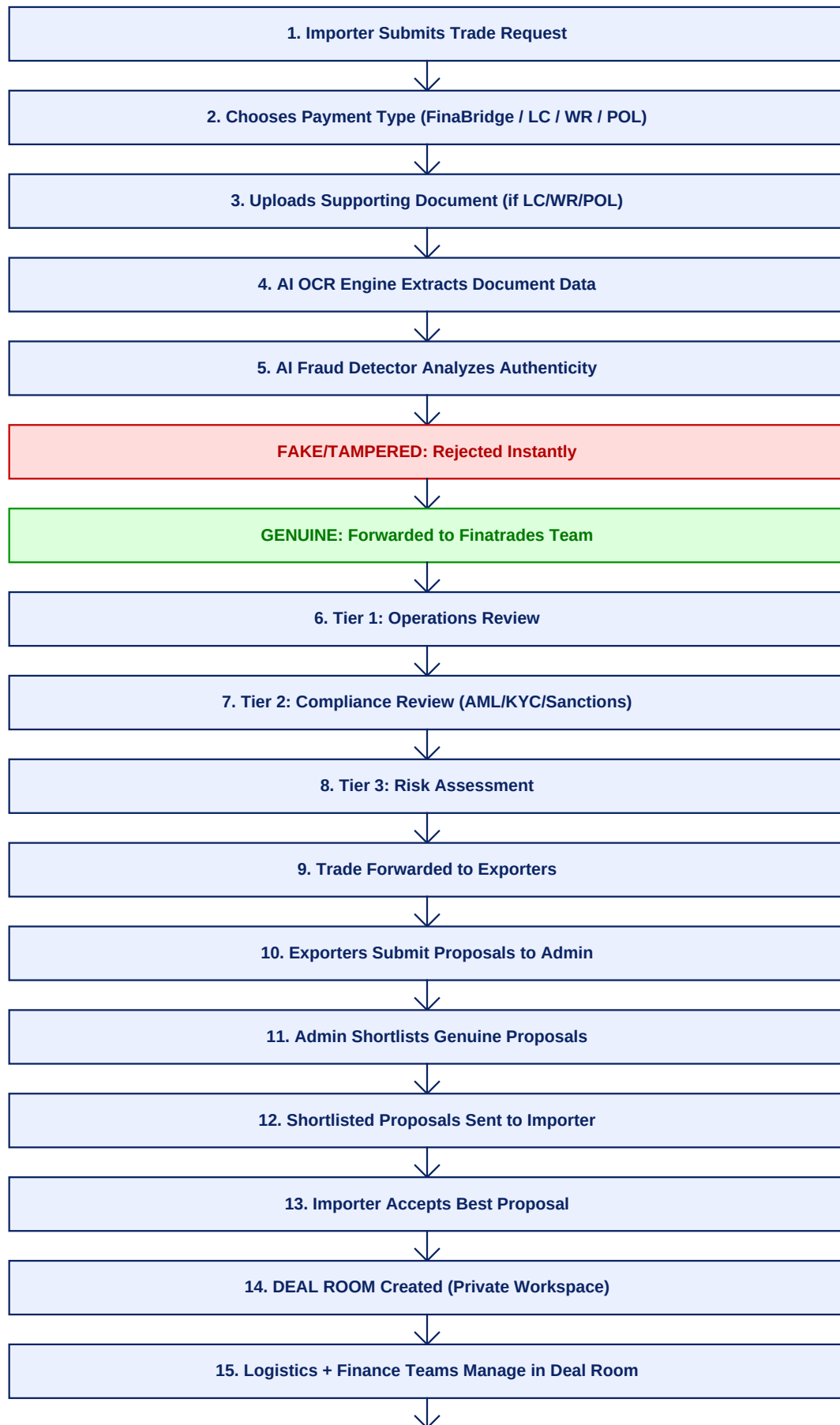
Option D: Facilitator Model WR Flow (IMPLEMENTED)



4. Complete Trade Lifecycle

The following diagram illustrates the complete trade lifecycle from initial submission to final settlement and certificate issuance. This represents the currently implemented Option D (Facilitator Model) flow, which is consistent across all payment types.

Complete Trade Lifecycle (Option D - Implemented)





5. AI Document Verification Pipeline

Every document uploaded to the Finatrades platform undergoes a rigorous AI-powered verification process. This automated pipeline serves as the first line of defense against fraudulent, tampered, or falsified trade documents. No fraudulent document can pass through this layer -- the AI detection system catches forgeries before any human review begins.

5.1 OCR Engine (Data Extraction)

- Automatically extracts all text, numbers, dates, and signatures from documents
- Identifies and parses key fields: issuing authority, amount, parties, terms
- Multi-language document support (English, Arabic, French, Chinese)
- Handwritten text recognition for signed documents
- Structured data output for downstream analysis

5.2 AI Fraud Detector (Authenticity Analysis)

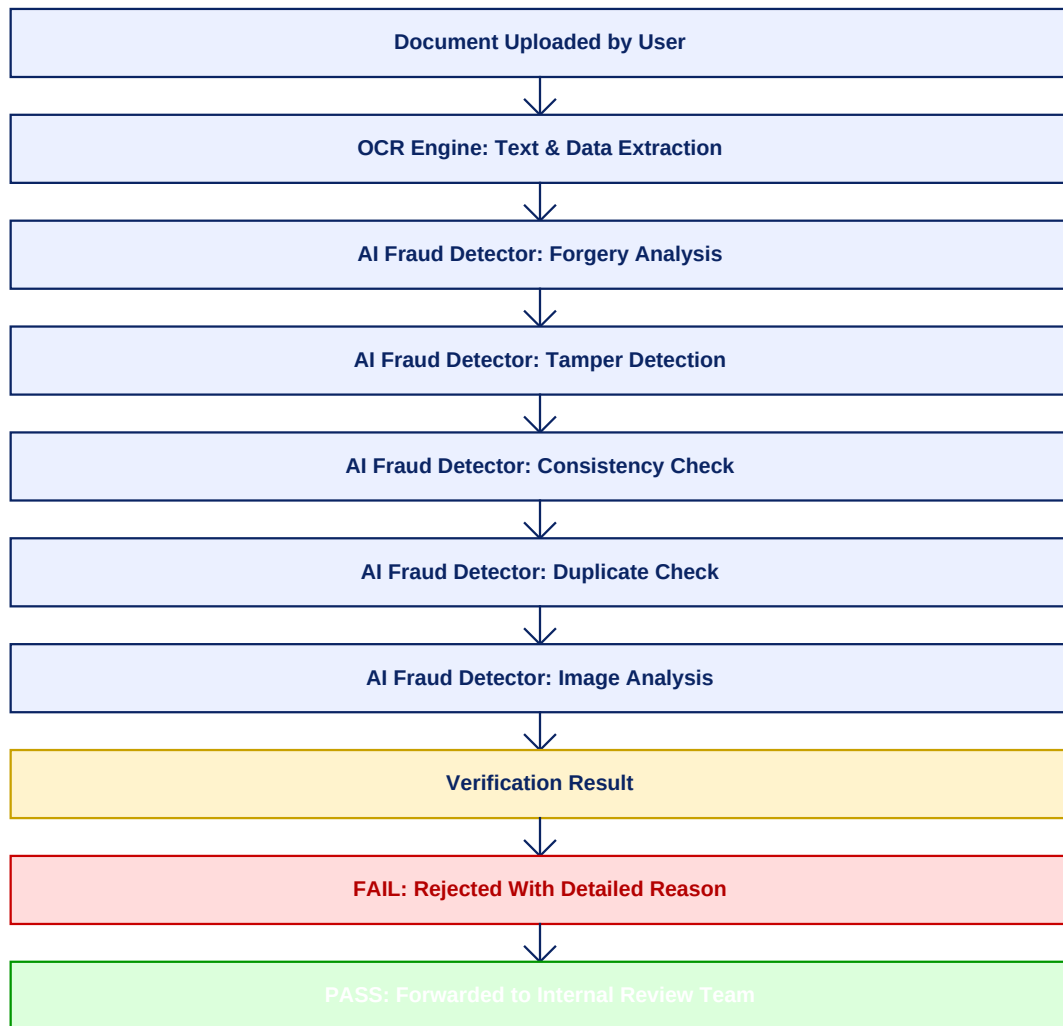
- **Forgery Detection:** Identifies digitally created or photocopied fake documents
- **Tamper Detection:** Detects pixel-level image manipulation, altered text, modified dates
- **Consistency Check:** Cross-validates data across document fields for contradictions
- **Duplicate Detection:** Checks if the same document has been submitted before
- **Image Analysis:** Examines document metadata, compression artifacts, font consistency
- **Authority Verification:** Validates bank codes, warehouse IDs, shipping company references

5.3 Verification Results

Each document receives one of two outcomes:

- **PASS:** Document is genuine, data is consistent, no anomalies detected. The document is forwarded to the internal review team with extracted data.
- **FAIL:** Document shows signs of forgery, tampering, inconsistency, or duplication. The submission is rejected immediately with a detailed reason. The user must re-upload a valid document.

AI Document Verification Pipeline



6. Internal Review Process (3-Tier)

After AI verification, documents enter the Finatrades internal review pipeline. This three-tier process ensures comprehensive human oversight covering operations, regulatory compliance, and financial risk assessment.

Tier 1: Operations Review

- Document completeness verification (all required fields present)
- Trade details accuracy check (commodity, quantity, value)
- Counterparty identification and validation
- Payment terms review and consistency
- Shipping/storage logistics feasibility assessment

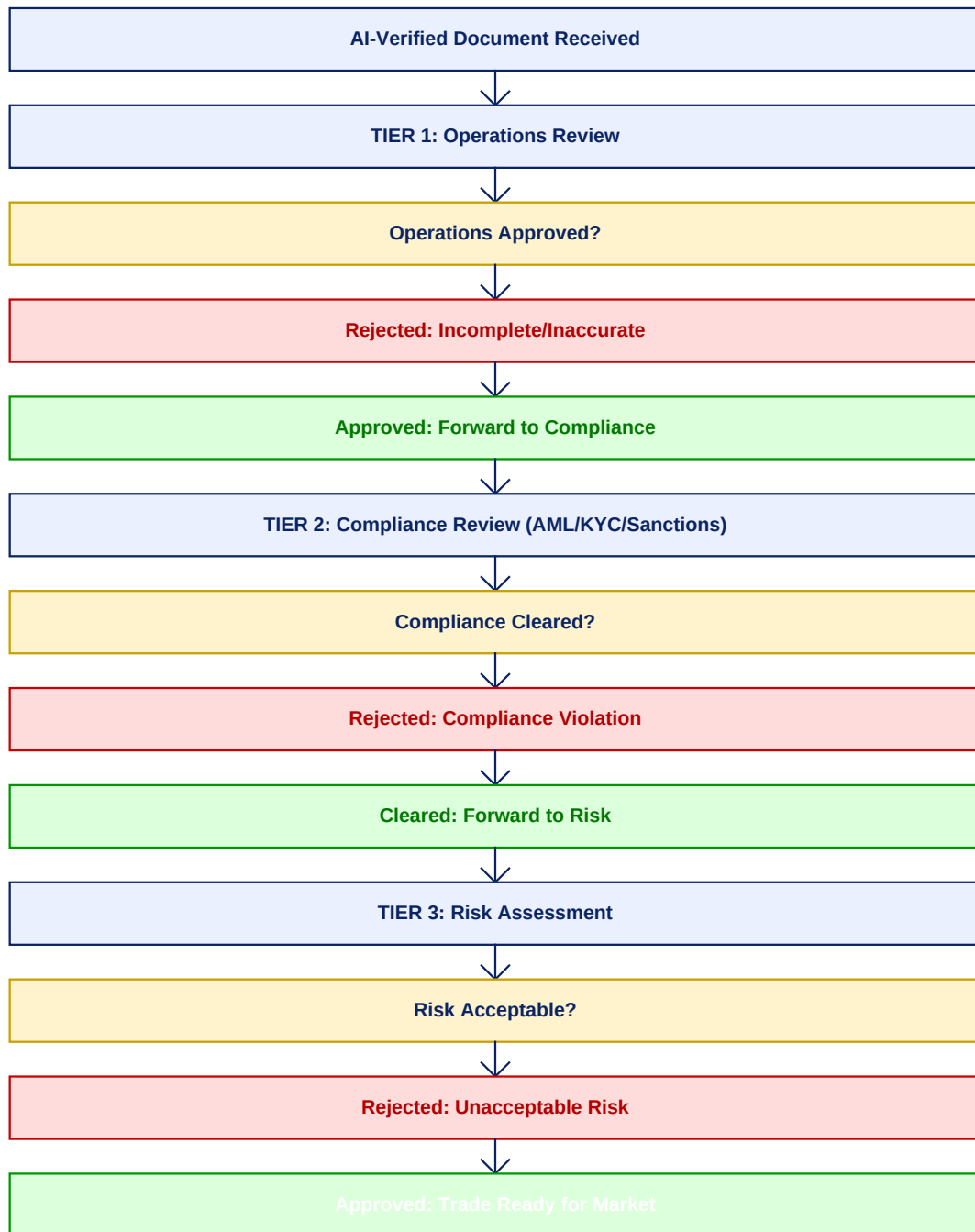
Tier 2: Compliance Review

- Anti-Money Laundering (AML) screening
- Know Your Customer (KYC) verification for all parties
- Sanctions list screening (OFAC, EU, UN, local)
- Swiss FINMA regulatory compliance check
- DIFC guidelines adherence verification
- Counter-Financing of Terrorism (CFT) screening

Tier 3: Risk Assessment

- Counterparty credit risk evaluation
- Trade value versus collateral ratio analysis
- Country and region risk scoring
- Market exposure and commodity price risk assessment
- Currency risk evaluation (if applicable)
- Historical trade performance review

3-Tier Internal Review Process



7. Proposal & Deal Room Flow

Once a trade passes all three tiers of internal review, it enters the proposal and matching phase. This process ensures that the importer receives only vetted, genuine proposals from qualified exporters, with administrative oversight at every stage.

7.1 Exporter Proposal Process

- Approved trade is published to qualified exporters on the platform
- Multiple exporters can view trade details and requirements
- Interested exporters submit detailed proposals to Admin
- Proposals include pricing, terms, timeline, shipping arrangements, and conditions
- Admin receives all proposals for review and quality assessment

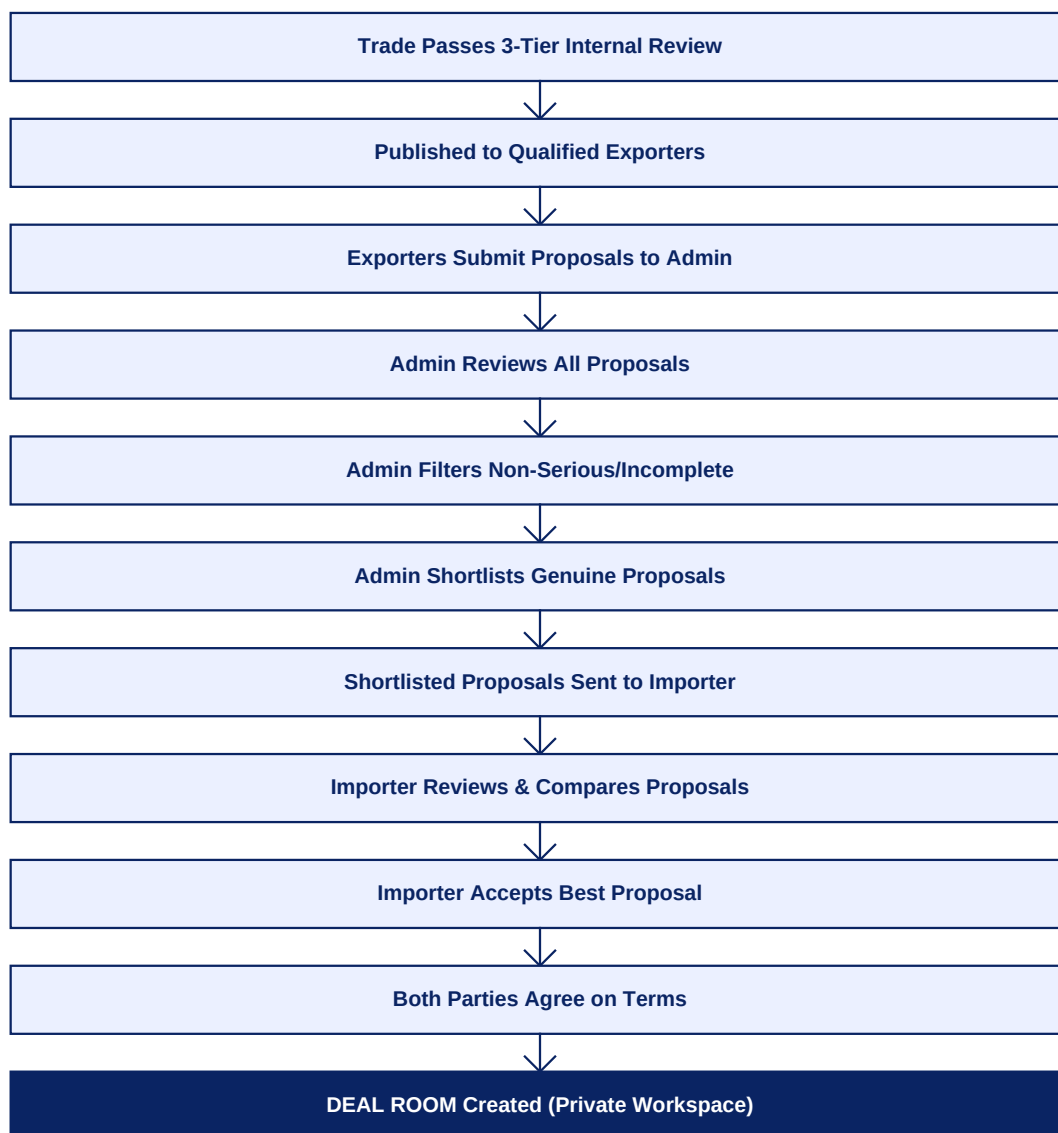
7.2 Admin Shortlisting

- Admin reviews all received proposals for completeness and quality
- Non-serious, incomplete, or overpriced proposals are filtered out
- Exporter credentials and track record verified
- Shortlisted genuine proposals are forwarded to the importer
- Admin provides summary notes on each shortlisted proposal

7.3 Importer Selection

- Importer reviews all shortlisted proposals
- Compares pricing, terms, delivery timelines, and exporter ratings
- Selects the best proposal based on their requirements
- Both parties formally agree on terms
- Acceptance triggers Deal Room creation

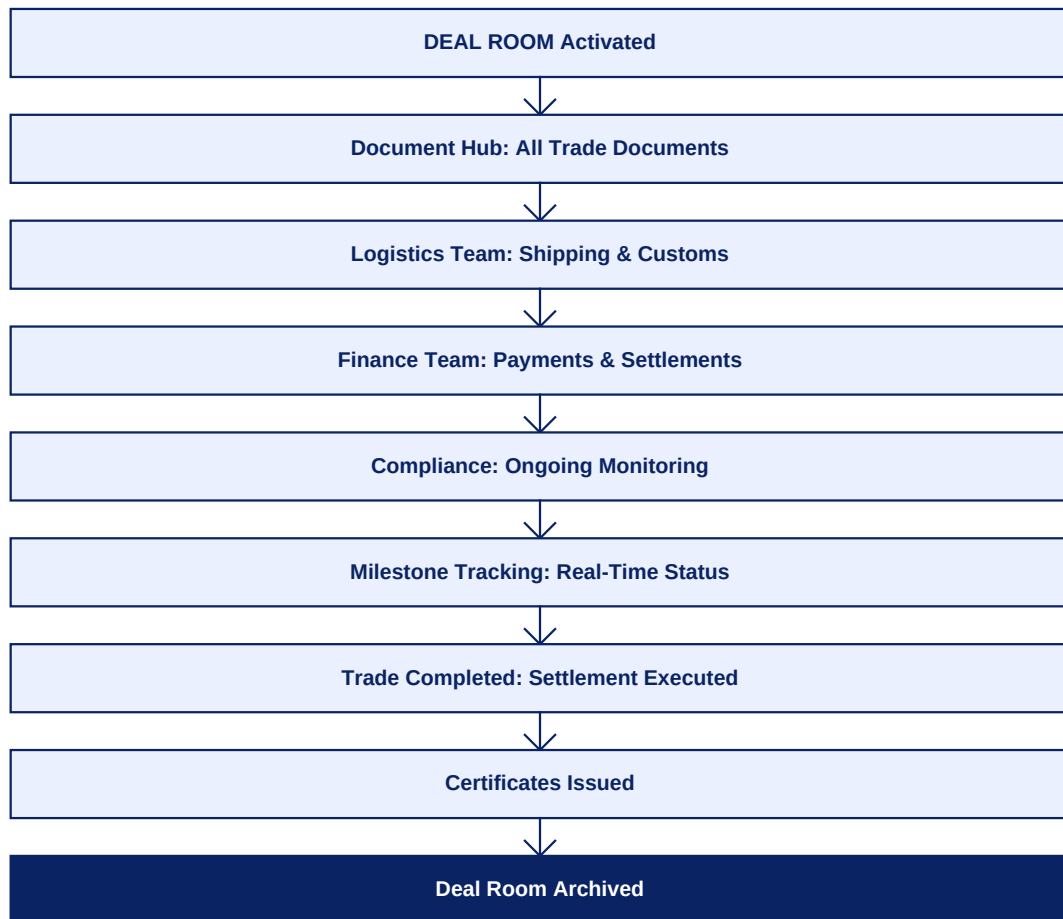
Proposal & Matching Flow



7.4 Deal Room Features

- **Document Hub:** All trade documents in one centralized location
- **Communication:** Secure messaging between importer, exporter, and internal teams
- **Milestone Tracking:** Real-time trade progress with automated status updates
- **Logistics Updates:** Shipping, customs clearance, and delivery tracking
- **Finance Dashboard:** Payments, credits, settlements, and FX management
- **Certificates:** Auto-generated trade completion certificates
- **Audit Trail:** Complete history of all actions, communications, and decisions

Deal Room Lifecycle



8. Security Architecture (5-Layer)

Finatrades employs a comprehensive 5-layer security architecture that ensures no fraudulent trade can progress through the system. Each layer provides independent verification, creating multiple barriers against fraud, non-compliance, and risk.

Layer 1: AI OCR + Fraud Detection (Automated)

AI-powered document verification catches forged, tampered, and fake documents before any human review begins. This is the fastest and most consistent layer.

Layer 2: Operations Review (Human)

Operations team verifies document completeness, trade accuracy, and counterparty legitimacy. Ensures all business requirements are met.

Layer 3: Compliance Check (Regulatory)

Full AML/KYC/CFT screening with sanctions list checks (OFAC, EU, UN). Ensures FINMA and DIFC regulatory compliance for every transaction.

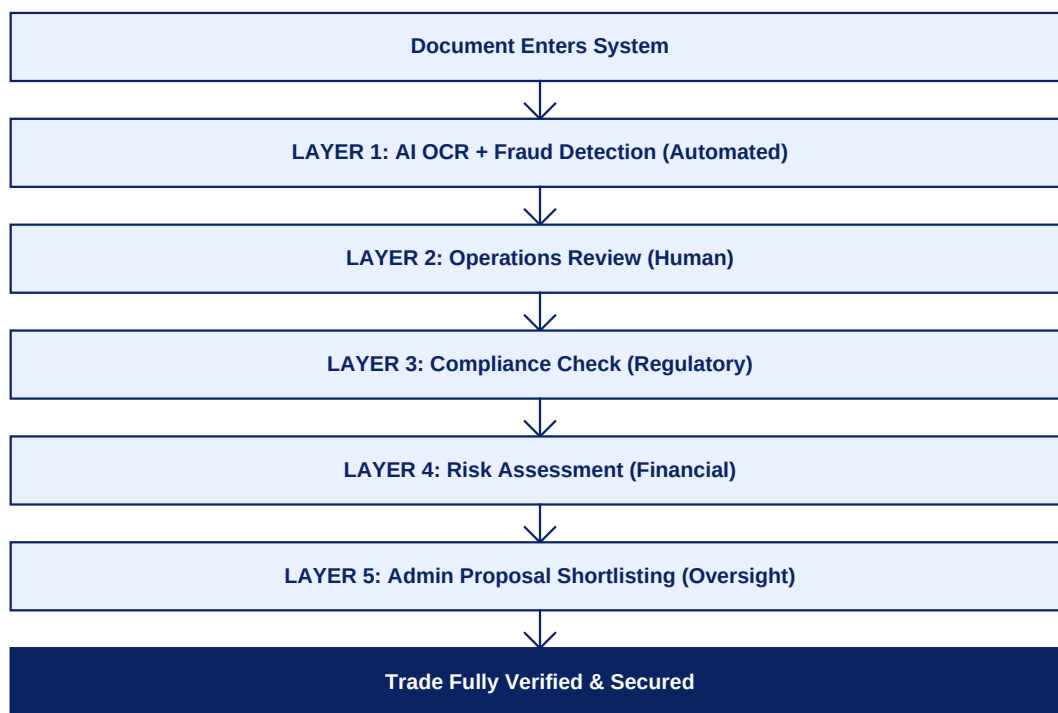
Layer 4: Risk Assessment (Financial)

Comprehensive credit risk, market risk, country risk, and collateral ratio analysis. Protects against financial exposure.

Layer 5: Admin Proposal Shortlisting (Oversight)

Final human filter ensuring only genuine, qualified exporter proposals reach the importer. Prevents low-quality or fraudulent counterparties.

5-Layer Security Architecture



9. Revenue Model

Finatrades generates revenue through a diversified fee structure across all trade finance instruments. As a facilitator, revenue is earned through transaction fees, interest spreads, and service charges without taking on direct counterparty risk.

Revenue Stream	Fee Range	Type	Basis
LC Facilitation	1-3%	Commission	LC Face Value
LC Credit Interest	12-18% APR	Interest	Credit Extended
POL Transaction	1.5-3%	Commission	Shipment Value
POL Credit Interest	10-15% APR	Interest	Credit Extended
WR Finance	2-4%	Spread	Commodity Value
WR Storage Fee	0.5-1%	Service	Annual Basis
Invoice Factoring	2-5%	Discount	Invoice Value
Supply Chain Finance	200-400 bps	Spread	Facility Size
Payment Guarantee	1-2%	Premium	Guaranteed Amount
Deal Room Usage	Fixed Fee	Service	Per Trade
FinaBridge Wallet	0.1-0.5%	Transaction	Transfer Amount

Revenue Distribution

- **Primary Revenue (60-70%):** LC, POL, and WR facilitation commissions and interest
- **Secondary Revenue (20-25%):** Supply chain finance spreads and invoice factoring
- **Platform Revenue (10-15%):** Deal Room fees, wallet transactions, premium services

10. Our Role: Pure Facilitator

Finatrades operates exclusively as a technology-driven facilitator in the trade finance ecosystem. This model eliminates direct risk exposure while maximizing value delivery to all participants in the trade lifecycle.

What We Do

- Provide AI-powered technology platform for trade document verification
- Offer secure Deal Room infrastructure for trade management
- Facilitate matching between importers and verified exporters
- Deliver 3-tier compliance and risk review services
- Enable gold-backed trade finance instruments
- Maintain 5-layer security architecture

What We Do NOT Do

- We do NOT hold gold directly (FinaVault partners handle custody)
- We do NOT execute trades (Deal Room facilitates, parties execute)
- We do NOT assume counterparty risk (AI + Compliance verifies)
- We do NOT provide direct lending (We facilitate credit through instruments)
- We do NOT act as a bank (We are a regulated technology facilitator)

Our Competitive Advantages

- **Technology:** AI-powered verification eliminates 95%+ of fraudulent documents
- **Speed:** Option D reduces processing from weeks to hours
- **Security:** 5-layer architecture exceeds banking security standards
- **Transparency:** Deal Room provides full visibility to all parties
- **Compliance:** FINMA + DIFC dual regulatory framework
- **Gold-Backed:** Stable value foundation immune to currency volatility

Implementation Status Summary

Option D (Facilitator Model): **ACTIVE & IMPLEMENTED**

AI Document Verification: **ACTIVE**

5-Layer Security: **FULLY OPERATIONAL**

End of Document

Finatrades Technical Architecture Team | March 2026

For questions or further details, contact the Trade Finance Division.